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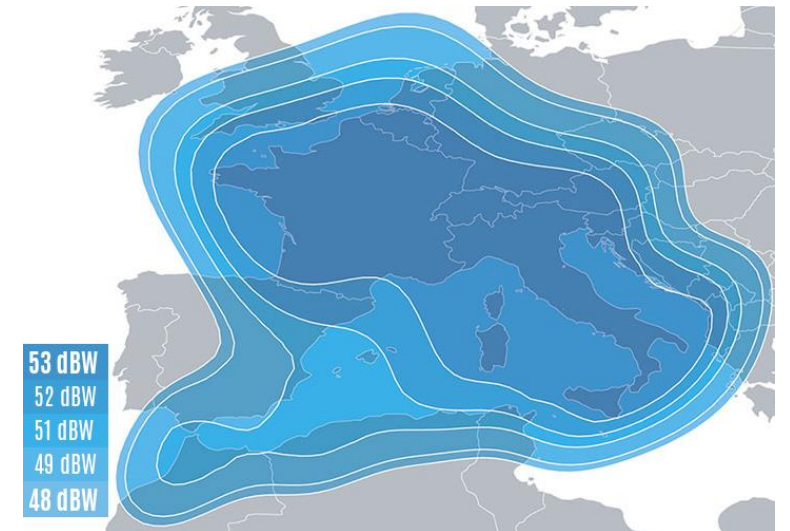
Team Eutelsat 5 West B

History of 5 West B (5WB)

- 5WB is a communications satellite in the 5° W slot in GEO
- Primarily operated by Fransat in the European/North African region
- 5WB was a successor to 5 West A (5WA), which also occupied the 5° W slot
- Shortly after leaving GTO, 5WB experienced a failure in one of the arrays
 - ❑ Reduced operational facilities to ~ 45%
- This required 5WA to stay in orbit for four more years



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Technical System Overview of 5WB

Parameter	Value
Launch Date	October 9, 2019
Launch Vehicle	Proton-M / Briz-M
Orbit Type	Geostationary (GEO)
Orbital Longitude	5° West
Orbital Period	1436.1 minutes
Inclination	0.10°
Satellite Launch Mass	2,864 kg
Satellite with Proton-M Launch Mass	705,000 kg
Payload	35 Ku-band transponders; EGNOS GEO-3
Satellite Bus	GEOSTar-2e
Design Lifetime	>15 years
Solar Array Power (nominal)	5500 W
Available Power (post-anomaly)	≈45% of nominal
Coverage Regions	Europe, North Africa
Cost	~100 million USD

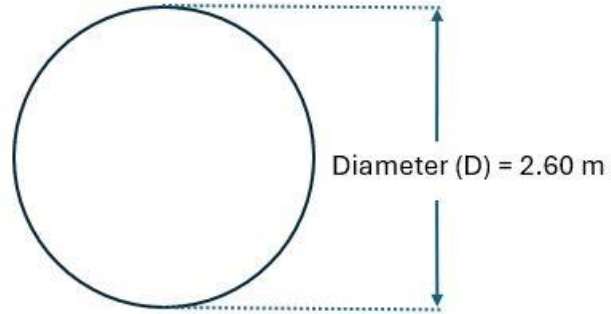
Link Budget Analysis

- Location: Pyrenees region in France (42.696°N, 2.406°E)
- $P_{r,up} = -105.4$ dBW
- $N_{up} = -128.6$ dBW
- $P_{r,down} = -122.6$ dBW
- $N_{down} = -136.5$ dBW
- $(C/N)_{down} = 13.9$ dB
- $(C/N)_{up} = 23.2$ dB
- $(C/N)_0 = 13.4$ dB
- $(C/N)_{min} = 7.9$ dB
- **Margin = 5.5 dB**

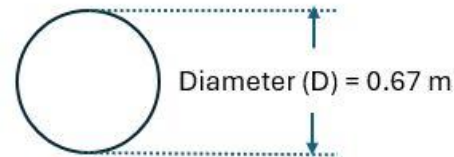
Parameter	Value
Earth Station Transmit EIRP ($EIRP_{up}$)	60 dBW
Satellite Transmit EIRP ($EIRP_{down}$)	53 dBW
Uplink Free Space Pathloss ($L_{0, up}$)	207.0 dB
Downlink Free Space Pathloss ($L_{0, down}$)	204.8 dB
Satellite Antenna Gain on the Uplink ($G_{sat, up}$)	49.4 dBi
Satellite Antenna Gain on the Downlink ($G_{sat, down}$)	47.3 dBi
Receive Antenna Gain (G_r)	35.2 dBi
Clear Air Loss on the Uplink ($L_{clear, up}$)	0.3 dB
Clear Air Loss on the Downlink ($L_{clear, down}$)	0.17 dB
Rain Loss on the Uplink ($L_{rain, up}$)	3.5 dB
Rain Loss on the Downlink ($L_{rain, down}$)	1.85 dB
Pointing Loss	3.0 dB
Miscellaneous Losses	1.0 dB

Link Budget Diagram

Inset Diagram of the Reflector Antenna on the satellite



Inset Diagram of the Reflector Antenna on the receiver



Satellite & Atmosphere

Uplink EIRP = 60 dBW
Physical air temp = 270 deg K
Satellite altitude = 36000 km

Design Criterion

Minimum overall C/N: 7.9 dB

Uplink (14.0 GHz)

Pointing loss = 3 dB
Path loss = 207.0 dB
Clean-air loss = 0.3 dB
Rain loss = 3.5 dB
Misc. losses = 1.0 dB

Downlink/Receiver (10.972 GHz)

Pointing loss = 3 dB
Path loss = 204.8 dB
Clean-air loss = 0.3 dB
Rain loss = 1.85 dB
System noise figure = 0.3 dB

GEO Orbit Path

Ground Station

Receiver

Additional Analysis I: Launch Site of 5WB in Baikonur Cosmodrome

- Baikonur Cosmodrome (first image) is located at (45° 57' 54" N, 63° 18' 18" E)
- This is unusual because the launch pad is located very far north for GEO satellites
 - Predominantly used for LEO satellites in polar orbit
- Reasons for this launch location:
 - Historical (i.e., the USSR's premier launch location)
 - Pre-existing infracture (been in use since 1955)
 - Cost and main launch for the Proton rocket



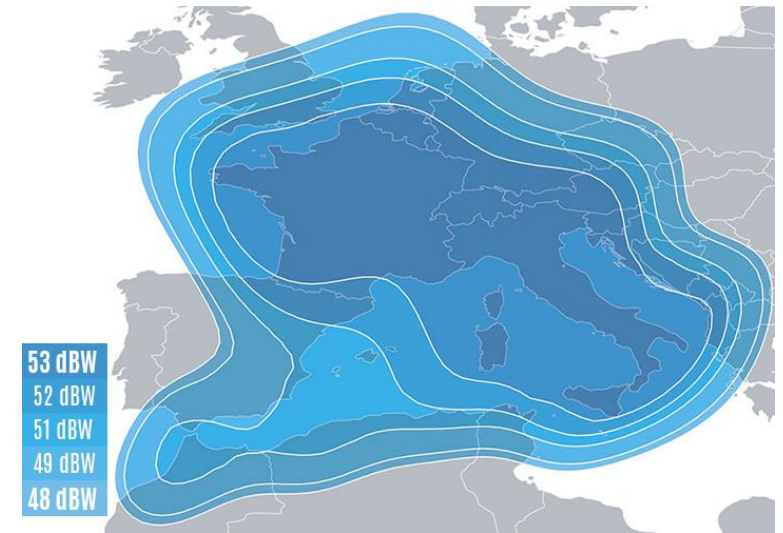
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Additional Analysis II: Coverage Case Study

- Three locations of interest:
 - Main teleport in Paris, France (48.858° N, 2.351° E)
 - Pyrenees region in France (42.696° N, 2.406° E)
 - Trondheim, Norway (63.430° N, 10.438° E)
- Main teleport: $(Az, EI) = (189.72^\circ, 33.48^\circ)$
- Pyrenees region: $(Az, EI) = (190.85^\circ, 40.15^\circ)$
- Trondheim: $(Az, EI) = (197.16^\circ, 17.23^\circ)$
 - Recall that the minimum EI for Ku-band is 20°
 - Most likely not get coverage despite the coverage map showing relatively high signal strength



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Future of 5WB

- 5WA was deployed with enough liquid fuel to last until 2034
- There are three possibilities for replacements for 5WB in the 5° W slot:
 - ❑ Like-for-like GEO replacement at 5° West
 - ❑ Consolidation into other Eutelsat GEO neighborhoods
 - E.g., HOTBIRD
 - ❑ Hybrid GEO/LEO services
 - As of 2023, Eutelsat merged with OneWebb, potentially gaining access to 35 GEO and 600 LEO satellites

Conclusions

- 5WB is a communications satellite in the 5° W slot in GEO
- A solar panel array limited the operational capacity of 5WB to 45%
 - The link budget example with the receive location in the Pyrenees region shows a margin of 2.3 dB
- The 5WB was launched from the Baikonur Cosmodrome
- Locations in France have a large elevation angle-well above the Ku-band limit
 - Areas in northern Europe, like Norway, have elevation angles that are very small
- 5WB is expected to be in service at least until 2034